

Leave the Sampler Alone: Truncation Changes Little Before a Model Collapses

Anonymous ACL submission

Abstract

Truncation samplers such as top- p , min- p , and top- $n\sigma$ exist to prevent the accuracy collapse of high-temperature sampling. Sampler evaluations compare one rule against another, mostly at temperatures where every model has already collapsed. We ask whether it occurs in every model at the temperatures deployments use, from the shipped defaults of 0.6 to 1.0 up to $T=1.3$. In a single controlled pipeline, we run seven instruction-tuned models from five families across 8 sampler arms, 4 quantization levels, and 2 tasks. At the shipped defaults, no truncation rule gains more than 3 points on any model outside the Llama family (8 at the 95% upper limit), and gains of 3 to 10 points appear only at $T=1.3$, on the three models with the largest temperature drops outside Llama. Under plain temperature sampling at $T=1.3$, Llama-3.1-8B and Llama-3.2-3B lose 34 to 41 accuracy points on GSM8K and MMLU-Pro, and Llama-3-8B loses 17 on MMLU-Pro; no model outside the family loses more than 8. The collapse is a termination failure. Generations derail midway and never reach an answer, but when a model does produce a well-formed answer, it is as accurate as under greedy decoding. A temperature ladder to $T=2.0$ on three models shows each collapsing at its own temperature, with truncation samplers mattering only past it. Forcing one off-distribution token into the context multiplies a fragile model’s chance of drawing another, so errors compound. Practitioners can screen a model for the collapse with fifty items and, if it passes, leave the sampler alone.

1 Introduction

Sampling is the default decoding choice for deployed language models (Shi et al., 2024), and truncation samplers exist to make it safe at high temperature. They were designed for open-ended generation, where output variety is the goal (Garces Arias et al., 2024), but they are now also evaluated on

tasks with one checkable answer. Min- p reports accuracy gains over top- p on GSM8K at high temperature (Nguyen et al., 2025), and top- $n\sigma$ reports accuracy maintained up to $T=3.0$ (Tang et al., 2024). Both results rest on the same premise, that raising the temperature eventually ruins every model’s accuracy and a good sampler limits the damage. We test that premise at the temperatures deployments actually use and ask **which models collapse in that range, and whether the sampler changes anything before they do.**

There is a good reason to expect the collapse. At each step the decoder draws the next token at random from the model’s predicted distribution, and a temperature parameter scales the randomness of the draw. Raising it moves probability from the model’s preferred tokens toward the rare ones in the tail of the distribution, where the model’s probability estimates are least reliable, so a high-temperature draw can select a token that fits the context poorly (Holtzman et al., 2020; Hewitt et al., 2022). The selected token then becomes context for every later step, and the continuation can degenerate into incoherent text. Truncation samplers prevent this degeneration by deleting the tail of the distribution before the draw, and the rules differ in where they cut.

The accuracy evidence has two problems. First, it generalizes from very few models. Top- $n\sigma$ ’s results come from a single model, Llama-3-8B (Tang et al., 2024), and min- p ’s main results from Mistral-7B (Nguyen et al., 2025). Second, the gains are demonstrated at temperatures of 1.5 to 3, while deployed systems ship defaults between 0.6 and 1.0 (Section 3).

Across nine instruction-tuned models from five families, run through one controlled pipeline, only one family collapses in that range, and outside it the sampler is worth at most a few points. The six models outside the Llama family hold their accuracy at deployment temperatures ($T \leq 1.3$) under

every sampler we test, staying within 8 points of their low-temperature scores. Llama-3.1-8B and Llama-3.2-3B lose 34 to 41 accuracy points under plain temperature sampling, and Llama-3-8B, top- $n\sigma$'s test model, loses 17 on the longer task.

Our contributions:

- **A single-pipeline factorial study.** Seven instruction-tuned models, four GGUF quantization levels plus a BF16 anchor, eight decoding arms, and $T \in \{0.7, 1.0, 1.3\}$, on GSM8K and MMLU-Pro ($N=50$ items, $K=3$ repetitions): 185,600 generations, with item-clustered bootstrap CIs on every headline estimate.
- **Robustness is model-bound, and the fragility runs in one family.** Every collapse we observe inside the deployment band is a Llama: two generations collapse outright and a third loses 17 points on the longer task (Section 4.1); on the robust models no truncation rule gains more than 3 points at the shipped defaults (8 at the 95% upper limit) (Section 4.2).
- **The collapse is a termination failure.** Among Llama's generations that reach a well-formed answer at $T=1.3$, accuracy matches greedy decoding on the same items. What changes is the probability of derailing into unparseable text mid-generation (Section 4.3).
- **A cliff map from $T=1.3$ to 2.0.** All three models we push to $T=2.0$ collapse, each at its own temperature. On GSM8K, Llama collapses by $T=1.5$, Qwen2.5 by 1.7, and Gemma-3 by 2.0. Truncation shifts each cliff right, so the gains the sampler papers report sit past it (Section 4.4).
- **A mechanism signature.** A logit probe and a forced-token experiment locate the fragility after the first off-distribution token enters the context. One injected tail token multiplies Llama's chance of drawing another 2.3-fold, while no robust model's rises by more than +0.0015 per position, about a third of Llama's increase (Section 4.6).

2 Related work

Truncation samplers. Top- k sampling draws from only the k most probable tokens; it was introduced for story generation, where likelihood-

maximizing decoding produced generic and repetitive text (Fan et al., 2018). Holtzman et al. (2020) showed that a fixed k fits poorly because the next-token distribution is sometimes peaked and sometimes flat, and proposed nucleus sampling, which keeps the smallest set of tokens whose probabilities sum to a target p . Min- p keeps only the tokens whose probability is at least a set fraction of the top token's, so the cut is tight where the model is confident and loose where it is uncertain (Nguyen et al., 2025). Top- $n\sigma$ cuts in logit space instead, keeping tokens within n standard deviations of the top logit; temperature rescales all logits equally, so the surviving set does not change with temperature (Tang et al., 2024). Further refinements of the cut exist, among them typical sampling and epsilon and eta sampling (Meister et al., 2023; Hewitt et al., 2022). Welleck et al. (2020) proved that top- k and nucleus decoding can produce sequences that never terminate even when the model itself always terminates, so non-termination is a documented decoding failure. In contrast to the sampler papers, which compare truncation rules against each other, we test whether the failure they all guard against occurs at deployment temperatures at all.

Reproducibility of sampler gains. Schaeffer et al. (2025) reanalyzed min- p on nine models and 31 temperatures, giving every sampler an equal number of hyperparameter settings, and found its reported gains "largely indistinguishable" from the other samplers'. A second variable in such a comparison is the order in which temperature and truncation are applied. The order matters because the truncation threshold is computed on whichever distribution the rule sees: applied before temperature, the rule cuts the model's original, sharp distribution; applied after, it cuts one that temperature has already flattened, so the same nominal setting keeps a different set of tokens. We run the two chain orders as explicit arms (Section 3). Garcés Arias et al. (2024) sweeps decoding hyperparameters on open-ended generation with quality and diversity metrics, while our tasks have one checkable answer. Shi et al. (2024) compare decoding methods on closed-ended tasks and report that the best method changes with the task and the model, and that quantization alters how robust a model is to the choice; the comparison is again among methods, and it leaves our premise question open. Song et al. (2025) documents per-model gaps between greedy decoding and sampling on reasoning benchmarks

and treats the gaps as evaluation noise to report; we treat the gap as the object of study and trace it to a termination failure in one model family.

Robustness to decoding randomness. Renze and Guven (2024) sampled temperatures from 0.0 to 1.6 on multiple-choice problem solving and found no statistically significant effect on accuracy between 0.0 and 1.0. We replicate that null for most models, but per-model results show Llama-3.1-8B falling just past $T=1.0$ while every other model holds (Section 4.1). The nearest study is Grover (2026), which also asks what makes a model robust to decoding randomness and concludes that instruction tuning is the primary determinant. Our data refine that conclusion. All nine of our models are instruction-tuned and one family still collapses, so instruction tuning cannot be the whole answer. Their evaluation differs from ours in two respects that matter. Their sweep stops at $T=1.0$, and the Llama collapse happens between 1.0 and 1.3; and their robust models score 11 to 30% on synthetic tasks, so there is little accuracy left to lose. They also test no truncation samplers. Fastowski et al. (2025) connect token-distribution entropy to temperature sensitivity on factual question answering; our logit probe supports the link (Section 4.6), though size does not decide the outcome in our grid, where Qwen3-1.7B holds and the larger Llama-3.1-8B collapses.

3 Experimental setup

The study is a factorial grid over four axes, model, quantization level, decoding arm, and temperature, crossed with two tasks. Table 1 lists the grid and the smaller experiments around it. We call one model at one quantization level, on one task at one temperature, a *cell*; the eight decoding arms run inside every cell on the same 50 frozen items. Every sampler comparison in the paper therefore holds the cell fixed and varies the arm, and every temperature comparison holds the arm fixed and varies only the temperature. Everything outside the axes is pinned: one llama.cpp commit, each model’s official chat template rendered client-side, the native completion API, and per-record seeds derived deterministically from the cell and arm. An accuracy difference between two runs is then attributable to the axes, up to sampling noise that the bootstrap below quantifies.

Decoding. Each cell runs eight arms: a deterministic greedy anchor, plain temperature sampling,

and the four truncation rules of Section 2, each at the setting its source paper uses or recommends: top- p 0.95 (Holtzman et al., 2020), top- k 40 (Radford et al., 2019), min- p 0.05 (Nguyen et al., 2025), top- $n\sigma$ 1.0 (Tang et al., 2024). Because the order of temperature and truncation changes which tokens survive (Section 2), we pin temperature first across the truncation arms and add two ablation arms that apply it last (the llama.cpp default) for top- p and min- p , the two rules whose surviving set depends on that order. Temperatures $T \in \{0.7, 1.0, 1.3\}$ span the deployment band. The defaults that models and engines ship sit between 0.6 and 1.0 (Llama-3.1’s generation config sets 0.6, llama.cpp defaults to 0.8, the OpenAI and Anthropic APIs to 1.0),¹ so 0.7 and 1.0 bracket common practice and 1.3 sits just above it. Every stochastic arm runs $K=3$ repetitions per item; greedy decoding is deterministic, so $K=1$.

Tasks. We evaluate GSM8K, grade-school math word problems (Cobbe et al., 2021) (0-shot chain-of-thought with the prompt of Meta’s Llama evaluation suite (Meta AI, 2024c), numeric exact match), and MMLU-Pro, ten-option multiple-choice questions across 14 categories (Wang et al., 2024) (0-shot chain-of-thought, letter exact match), $N=50$ items each, a 512-token budget on GSM8K and 1024 on MMLU-Pro, with parsers faithful to Im-eval-harness (Gao et al., 2023). Both tasks require a chain of reasoning ending in one exactly checkable answer, which keeps grading free of judge noise and leaves a long generation in which a derailment can occur. The two tasks differ in generation length, with MMLU-Pro’s chains running roughly twice as long as GSM8K’s, a contrast Section 4.4 uses to test whether longer generations fail earlier. Both tasks grade a generation by parsing a final answer out of the generated text, so the parser decides how a generation that never reaches an answer is graded. We log the parse path of every record: strict when the pinned final-answer sentence is present (“The final answer is 26”; “The answer is (I)”), flexible when it is absent and the parser falls back to the last number (GSM8K) or the last parenthesized option letter (MMLU-Pro), and failed when neither exists, which grades as incorrect. Appendix B shows one generation per path and checks the rule against markdown formatting. Every contrast in the study compares runs on the same frozen items.

¹Sources: Llama-3.1-8B-Instruct’s generation_config.json, llama.cpp’s defaults, the OpenAI and Anthropic API references (July 2026).

Table 1: The experiments. Both tasks run in every row except Stratified (MMLU-Pro only, fresh items); the BF16 anchor is Qwen3-4B only; the arms are listed under Decoding. Gen. counts graded generations, or prompts and continuations for the two probes (greedy runs once per item; two Mistral prompts end before the injection point and are skipped).

Experiment	Models	Quantization	Arms	T	$N \times K$	Gen.	Where
Main grid	7 (5 families)	Q8/Q6/Q4/Q3 (+BF16)	8	0.7, 1.0, 1.3	50×3	185,600	\$4.1–4.5
Cliff ladder	Llama-3.1, Qwen2.5, Gemma-3	Q8	4	1.3, 1.5, 1.7, 2.0	50×3	14,400	\$4.4
Lineage	Llama-3-8B, Llama-3.2-3B	Q8	greedy, temp.	0.7, 1.0, 1.3	50×3	2,000	\$4.1
Stratified	Llama-3.1, Qwen2.5, Gemma-3	Q8	greedy, temp.	0.7, 1.0, 1.3	50×3	1,500	App. C
High T	Llama-3.1, Gemma-3	Q8	4	2.0, 3.0	15×3	1,440	App. E
Thinking on	Qwen3-8B	Q8	3	0.7, 1.3	25×2	450	App. G
Logit probe	7	Q8	greedy	–	50×1	700	\$4.6
Forced token	7	Q8	greedy, inject.	–	50×1	1,396	\$4.6

Models and quantization. The grid holds seven instruction-tuned models from five model families: Llama-3.1-8B (Grattafiori et al., 2024), Mistral-7B-v0.3 (Jiang et al., 2023), Qwen2.5-7B (Yang et al., 2024), Qwen3-1.7B/4B/8B (Yang et al., 2025) (thinking off; a thinking-on spot-check in Appendix G), and Gemma-3-12B (Gemma Team, 2025). A follow-up temperature-only run adds two more Llama generations, Llama-3-8B (Meta AI, 2024a) and Llama-3.2-3B (Meta AI, 2024b) (Section 4.1). Each model runs at four quantization levels (Q8_0, Q6_K, Q4_K_M, Q3_K_M), all imatrix GGUF conversions from a single provider so the quantization axis is not confounded by the conversion recipe. A BF16 anchor on Qwen3-4B matches Q8 within noise, which lets Q8 stand in for full precision.

Statistical inference. All confidence intervals come from a nonparametric bootstrap ($B=2000$; 5000 for the contrasts of Appendix K) clustered on task items: we resample the 50 items rather than individual generations, because the $K=3$ repetitions of one item share that item’s difficulty and treating them as independent would understate the uncertainty. Where a comparison shares items across its two sides, which is everywhere in the grid, the bootstrap is paired.

Hardware and reproducibility. The study totals about 205,000 graded generations (Table 1). Hardware and the pinned llama.cpp commit are in Appendix L; model-file hashes are in the repository’s DOWNLOADS.md.

4 Results

4.1 The collapse is model-bound

We begin with plain temperature sampling and no truncation, because the accuracy a model loses when its full distribution is sampled at a higher

Table 2: Plain-temperature accuracy drop $T=0.7 \rightarrow T=1.3$ (pp, 95% item-clustered bootstrap CIs; pooled over quantization levels, five for Qwen3-4B including BF16) and sampler max–min accuracy spread at $T=1.3$ (pp; CIs in Figure 4).

Model	Task	drop [95% CI]	spread
Llama-3.1-8B	GSM8K	+34.0 [28.5, 39.7]	36.7
	MMLU-Pro	+40.8 [31.8, 50.3]	41.8
Mistral-7B-v0.3	GSM8K	+3.5 [−1.2, 8.5]	5.3
	MMLU-Pro	+8.0 [3.5, 13.2]	10.0
Qwen2.5-7B	GSM8K	+4.0 [1.2, 7.3]	3.8
	MMLU-Pro	+4.8 [1.3, 8.8]	5.7
Gemma-3-12B	GSM8K	+2.0 [−0.0, 4.8]	3.3
	MMLU-Pro	+3.7 [0.2, 7.7]	5.8
Qwen3-8B	GSM8K	+0.5 [−1.5, 2.6]	1.8
	MMLU-Pro	+1.0 [−2.4, 4.8]	3.5
Qwen3-4B	GSM8K	+2.1 [0.4, 3.9]	1.3
	MMLU-Pro	+0.7 [−1.0, 2.5]	1.7
Qwen3-1.7B	GSM8K	−0.3 [−3.8, 3.0]	3.2
	MMLU-Pro	+0.7 [−3.4, 5.0]	5.5

temperature is exactly the failure the truncation samplers of Section 2 exist to prevent. Figure 1 plots the drop from $T=0.7$ to $T=1.3$, pooled over quantization levels, and Table 2 gives the exact values.

The models form a gradient rather than two groups: the Qwen3 family drops at most 2 pp, Qwen2.5, Gemma, and Mistral drop 2 to 8 pp with at least one task CI excluding zero, and Llama drops 34 to 41 pp. Llama’s drop begins inside the shipped band: from $T=0.7$ to 1.0 it loses 8 pp on MMLU-Pro and its run-to-cap rate doubles (Appendix B). Three controls rule out the obvious confounds. Accuracy headroom does not explain the collapse: Qwen3-1.7B scores 74% on GSM8K, close to Llama’s 78%, and does not collapse. Model age does not explain it either, since Mistral-7B-v0.3 predates Llama-3.1 and is robust. The stratified MMLU-Pro replication reproduces the pattern on fresh items, giving Llama +43.3 pp

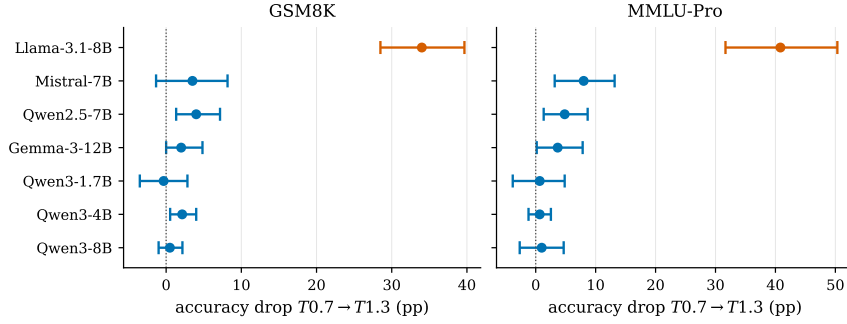


Figure 1: The pure-temperature collapse is model-bound: accuracy drop $T_{0.7} \rightarrow T_{1.3}$, points with 95% item-clustered bootstrap CIs.

[31.3, 55.3], Qwen2.5 +6.0, and Gemma +7.3.

A single collapsing checkpoint could still be a quirk of that release rather than a family trait, so a follow-up run repeats the pure-temperature axis on two more Llama generations, Llama-3-8B and Llama-3.2-3B (Q8, same frozen items, $K=3$; drops and accuracy levels in Appendix D). Llama-3.2-3B collapses like Llama-3.1-8B (+38.0 pp on GSM8K, +36.7 on MMLU-Pro) and with the same degeneration signature as Section 4.3. Llama-3-8B sits between the two: a null on GSM8K but +16.7 pp on MMLU-Pro, the longer task, a larger drop than any non-Llama model shows in Figure 1. Three Llama generations produce the three largest MMLU-Pro drops in the study. The fragility recurs across the family’s releases, and what sets it apart from the other four families is the open question of Section 5. One family collapses inside the deployment band, and no model outside it loses more than 8 points.

4.2 The sampler changes little before the cliff

This subsection holds temperature at $T=1.3$, the top of the deployment band, and varies the sampler. The maximum minus minimum accuracy across the seven stochastic sampler arms is 1.3 to 10.0 pp on every robust model (Table 2, spread column; per-arm accuracy grids in Appendix A), and even those numbers overstate the axis, because the maximum and minimum of seven noisy estimates drift apart by chance. On Llama the spread is not sampler noise: under plain temperature it has already collapsed at $T=1.3$ and under top- $n\sigma$ it has not (Section 4.4), so the spread measures how much of the collapse each rule prevents.

We therefore compare each of the six truncation arms with plain temperature directly: for each model, task, and temperature, the paired accu-

racy difference on the same items, with an item-clustered bootstrap CI (all contrasts in Appendix K; a GEE over the full grid, clustered on item, reproduces every interval to within 0.6 pp). At $T=0.7$ and 1.0, the temperatures models ship with, no arm gains more than 3.3 pp on any robust model, no pointwise upper 95% limit exceeds 7.5 pp (a simultaneous bound on the largest gain over all 144 contrasts is 7.8), 135 of 144 point estimates lie within 3 pp of plain temperature, and 6 intervals exclude zero (2 above, 4 below), the number chance alone gives. At $T=1.3$, 22 of 72 contrasts lie above zero, all on Mistral, Qwen2.5, and Gemma-3, the three non-Llama models with the largest temperature drops, with gains of 3 to 10 pp, largest on the longer task; no Qwen3 contrast leaves zero at any temperature. On the robust models truncation is worth nothing at the shipped defaults and a few points once a model has begun to collapse; on Llama the hardest-cutting rules are already worth 6 to 11 pp on MMLU-Pro at $T=1.0$ (Appendix K), where its collapse has begun.

Chain order changes accuracy by at most 4 pp on the robust models at $T \leq 1.0$ and by up to 7 pp on Mistral at $T=1.3$; on Llama, truncating before temperature scaling is more protective at $T=1.3$ (top- p 64% $\rightarrow$ 73% on GSM8K), as the threshold mechanics of Section 2 predict (both orders in Appendix F).

4.3 The collapse is degeneration, not wrong answers

This subsection works under plain temperature sampling throughout. An accuracy drop can come from two failures: completed answers that are wrong, or generations that never reach an answer. The parse path that Section 3 logs for every record separates the two, so we decompose Llama’s collapse along it

(Figure 2; full decomposition and example outputs in Appendix B).

Nearly all of the drop is the second failure. At $T=1.3$, 53.8% of Llama’s GSM8K generations and 89.2% of its MMLU-Pro generations derail into token-level noise and run to the token cap. Among the generations that do produce a well-formed final answer (a strict parse: the pinned final-answer sentence is present), Llama still scores 97.3% on GSM8K at $T=1.3$. That number needs one correction. Survival is selective, because easier items produce shorter generations and shorter generations have less room to derail, so the 97.3% is measured on an easier-than-average set of items. The right comparison for it is greedy decoding on those same surviving items at the same quantization level, which scores 97.3%, against 78.0% on all items. Among the generations that finish, then, accuracy at $T=1.3$ matches greedy decoding (97.3 vs. 97.3% on GSM8K; 76.9 vs. 78.8% on MMLU-Pro).

The two tasks report one failure differently: degenerate text almost always contains a number, so GSM8K’s last-number fallback grades a derailed generation as a wrong answer, while MMLU-Pro’s letter parser reports a parse failure. Temperature does not raise Llama’s rate of wrong answers; it raises the rate of generations that never reach one.

4.4 The cliff map: robustness is cliff location

To see whether the robust models also collapse at higher temperature, we extend the temperature axis to $T \in \{1.3, 1.5, 1.7, 2.0\}$ on three models spanning the gradient of Figure 1: Llama-3.1-8B, Qwen2.5-7B, and Gemma-3-12B (Q8, both tasks, $N=50$, $K=3$). Figure 3 and Table 3 give the trajectories and the drops against each model’s own $T=0.7$ anchor. Every model turns out to have a cliff, and what distinguishes the models is where it sits. Calling the cliff the first rung at which a model loses more than half its $T=0.7$ accuracy, the three models collapse on GSM8K in their Figure 1 order: Llama at 1.5, Qwen2.5 at 1.7, and Gemma at 2.0. At $T=1.5$ the CI of Llama’s drop does not overlap the other two models’, and at $T=1.7$ Qwen2.5’s does not overlap Gemma’s. Within each model the cliff sits earlier on MMLU-Pro, the task with roughly twice the generation length, as expected if the risk is per token and accumulates over a longer generation. There Llama has already collapsed at $T=1.3$, and Qwen2.5 and Gemma fall together at $T=1.5$.

Truncation shifts the cliff right in order of tail-

Table 3: The cliff ladder: accuracy drop against each model’s own $T=0.7$ anchor under plain temperature (pp, 95% item-clustered bootstrap CIs; Q8). The ladder is a separate run from the main grid; its $T=1.3$ rung differs from Table 2 by amounts within the CIs (Appendix L).

T	Llama-3.1-8B	Qwen2.5-7B	Gemma-3-12B
<i>GSM8K</i>			
1.3	+32.7 [24.7, 41.3]	+4.0 [−0.7, 9.3]	+2.7 [−2.0, 8.0]
1.5	+68.0 [58.0, 76.7]	+24.0 [15.3, 33.3]	+14.7 [8.7, 21.3]
1.7	+76.0 [66.7, 84.7]	+70.0 [62.0, 77.3]	+42.7 [34.7, 50.7]
2.0	+77.3 [67.3, 86.0]	+90.7 [82.7, 97.3]	+84.0 [76.7, 90.7]
<i>MMLU-Pro</i>			
1.3	+44.7 [34.0, 56.0]	+8.0 [0.0, 16.7]	+3.3 [−3.3, 10.0]
1.5	+46.0 [35.3, 57.3]	+40.0 [28.7, 52.0]	+39.3 [29.3, 48.7]
1.7	+48.0 [36.7, 60.0]	+61.3 [48.7, 74.0]	+57.3 [45.3, 68.0]
2.0	+48.0 [36.7, 59.3]	+65.3 [53.3, 78.0]	+63.3 [51.3, 75.3]

cutting strength: top- p moves the cliff about 0.2 higher, min- p holds 70% at $T=1.5$ on Llama GSM8K where plain temperature is at 9%, and top- $n\sigma$ is flat to $T=2.0$ on all three models. The strict-parse rate tracks accuracy almost one-to-one along the whole ladder, so the accuracy cliff coincides with the degeneration cliff. This reconciles the sampler papers with Section 4.2: min- p and top- $n\sigma$ are validated at temperatures of 1.5 to 3 (Nguyen et al., 2025; Tang et al., 2024), past every model’s deployment range, where their gains are real (Appendix E).

4.5 Quantization and answer agreement

The results so far pool over the four quantization levels. For that to be safe, quantization must not cost accuracy on its own, and it must not change what temperature and the samplers do. Quantization costs little accuracy. From Q8_0 down to Q3_K_M, greedy accuracy is unchanged on every model of 7B parameters or more, and a real cost appears only on the two smallest models, 7 pp on Qwen3-4B and 9 pp on Qwen3-1.7B. It changes nothing about the collapse either. Recomputed separately at each level, the plain-temperature drop of Section 4.1 shows Llama-3.1 collapsing at every level, with drops of 30 to 47 pp across levels and tasks, while no robust model exceeds 12 pp at any single level. Llama’s sampler ordering at $T=1.3$ is identical at all four levels (per-level values in Appendix H).

A second check separates robustness from determinism. A model with very peaked next-token distributions would also hold its accuracy, because sampling would return the greedy path every time and the three repetitions of an item would come

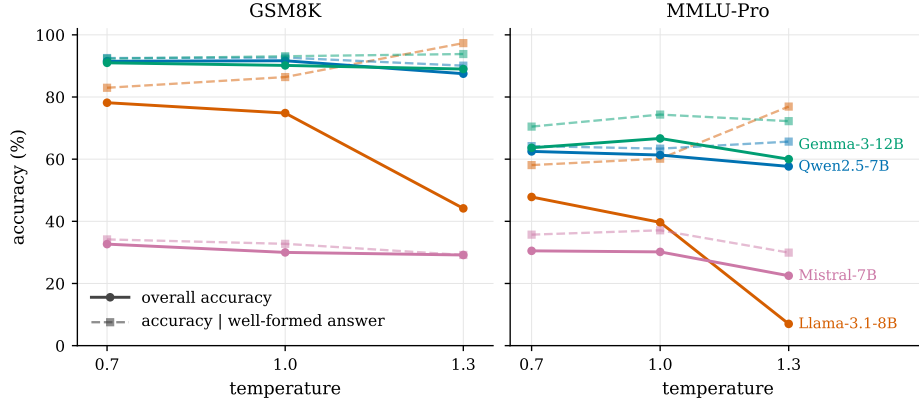


Figure 2: Overall accuracy (solid) versus accuracy among well-formed, strict-parsed answers (dashed) under plain temperature sampling; four models spanning the robustness gradient (the three Qwen3 models are flat and omitted for legibility). The dashed lines never fall: among well-formed answers accuracy holds, and Llama’s rise is survivorship (Section 4.3).

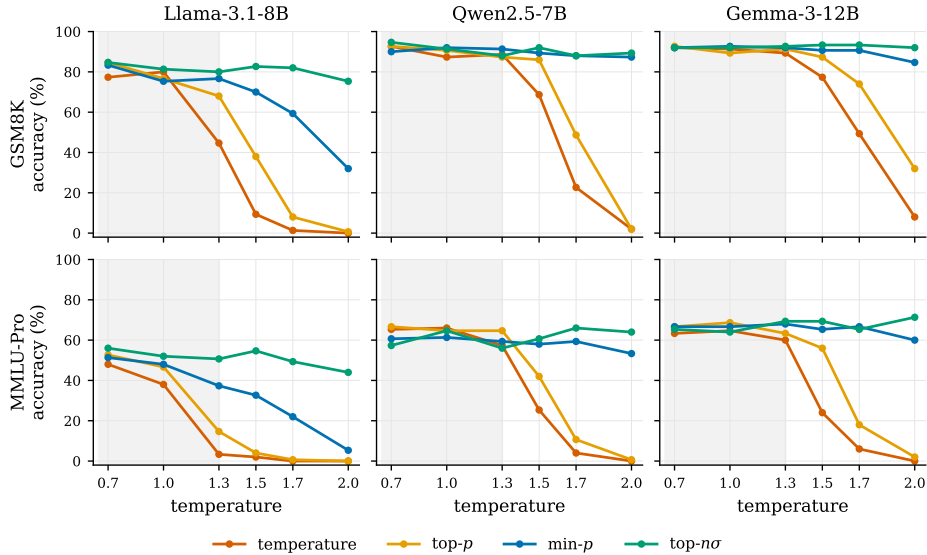


Figure 3: The cliff map. Accuracy versus temperature per sampler (the $T \leq 1.0$ points come from the main matrix; all at Q8). Each model has a cliff, and where it sits differs by model. Truncation shifts each cliff right.

out as near copies. The repetitions show the opposite. No item’s three repetitions are identical on any model, yet the robust models still converge on the answer, 1.1 to 1.4 distinct parsed answers per item on GSM8K at $T=1.3$ (Mistral, whose low GSM8K accuracy scatters its answers, is 2.5; Appendix I). Llama-3.1 rises from 1.5 to 2.3 under the same change, as each repetition derails on its own (full table in Appendix I). Neither quantization nor determinism explains the pattern.

4.6 Toward a mechanism: per-token uncertainty predicts the ordering of drops

If the collapse is a derailment process, fragile models should carry more per-token uncertainty before

any derailment happens. To measure that uncertainty we decode the 100 frozen prompts (the 50 items of each task) greedily on every model at Q8 and record the raw top-20 next-token probabilities at each position. Each prompt is summarized by four statistics of these distributions: an entropy lower bound (only the top 20 probabilities are observed), the top-1 probability and its margin over the top-2, and the fraction of positions where the top-1 probability falls below 0.5. We decode greedily so that the statistics describe the text each model produces when no random draw is involved, before any random draw can derail it.

Mean greedy-path entropy ranks the seven models’ temperature drops at Spearman $\rho = +0.93$

(flat-position fraction $+0.93$; top-1 margin -0.93 ; plotted in Appendix J). With seven models a rank correlation is fragile, so we report an item-clustered bootstrap interval, $[+0.54, +0.96]$, and Appendix J adds an exact permutation test ($p = 0.007$) and leave-one-model-out ranges. The summaries cannot separate one pair, though. Llama-3.1 and Mistral are nearly tied on every statistic we record, 0.33 versus 0.32 nats of entropy and 1.0×10^{-3} probability mass beyond the top 20 for both, yet Llama drops 40.8 pp on MMLU-Pro and Mistral 8.0. Uncertainty on the greedy path predicts which models drop most, but not by how much.

The probe also estimates the per-token risk that drives a derailment. For each position we compute the probability that sampling at $T=1.3$ picks a token from beyond the model’s own top 20; we call this the escape hazard (Appendix J states the estimator and its tail assumption). Llama’s is 0.0033 per position and Mistral’s 0.0034, about 0.85 expected tail draws per 256 positions for both, and across the seven models the hazard ranks the drops at the same $\rho = +0.93$ (bootstrap $[+0.57, +0.96]$). Generation lengths show that a derailed generation does not recover: on every robust model the length distribution of finished generations is unchanged from $T=0.7$ to $T=1.3$ (90th-percentile ratio 0.85 to 1.19), and Llama’s finished generations at $T=1.3$ are *shorter*, because only the short ones finish (Section 4.3). Llama and Mistral therefore draw tail tokens equally often, and a generation that derails does not come back, so what separates them must be how often a tail token starts a derailment, that is, what happens after that token is in the context.

A perturbation experiment measures that directly. For each model and frozen prompt we greedily-decode a 48-token prefix, replace the next token with the model’s own rank-20 candidate at that position (median raw probability 2×10^{-9} , a genuine tail draw), and continue greedily for 256 tokens, alongside a control continuation of the unedited prefix. We then compare the two continuations’ escape hazard over the 32 positions after the injection.

Beyond the injected token itself, the continuations show no visible damage. They stay coherent, and Llama’s reach the token cap no more often than its controls (14.0% for both on GSM8K). The difference is in the hazard. Llama’s rises to 2.3 times its baseline ($+0.0042$ per position), Mistral’s to 1.35 times ($+0.0012$), and no robust model gains

more than $+0.0015$ (the absolute increase, not the ratio; Appendix J explains why). This per-model increase ranks the temperature drops at $\rho = +0.89$ (bootstrap $[+0.39, +0.96]$, permutation $p = 0.012$; the entropy increase, $+0.86$). With seven models these are consistency checks, not a confirmed causal account. In short, Llama and Mistral draw tail tokens at the same rate, and what sets Llama apart is how much one tail token raises the chance of the next.

5 Discussion

For practitioners. At deployment temperatures the consequential choice is the model: on a robust one every configuration we test stays within 10 points and most within 3, on a fragile one the same sweep spans 42. Screening is cheap: fifty items under plain temperature at the top of the deployment band suffice, and the drop in strict-parse rate between $T=0.7$ and 1.3 identifies a fragile model. In production, a derailed generation carries no well-formed answer, so simple output validation catches it, while the rate of well-formed but wrong answers barely moves.

For the sampler literature. A sampler evaluation should say where its models’ cliffs sit, since a gain measured past one model’s cliff need not exist on a model whose cliff lies further out (Section 4.4).

6 Conclusion

Across nine models from five families, the choice of truncation sampler matters far less at deployment temperatures than the model’s own temperature sensitivity. Robust models hold within a few points under every sampler we test, while two Llama generations lose up to 41 points under plain temperature and a third loses 17 on the longer task. Each model we push past the deployment band collapses at its own temperature, Llama already inside the band; truncation raises that temperature but changes little below it. These results argue for screening temperature fragility before deployment rather than tuning the sampler after it. What sets a model’s cliff remains open.

Limitations

Our sample is 50 items per task with $K=3$ repetitions per cell; the small item count is what makes a factorial of 185,600 generations feasible. Item-clustered bootstrap CIs accompany every headline

estimate, and the headline drops are 2 to 4 times their CI widths. Mistral-7B’s GSM8K accuracy is low (32.7% at $T=0.7$), which limits the power to detect a further drop; its MMLU-Pro drop is measured away from floor and its CI excludes zero. The frozen MMLU-Pro subset is single-category (business) because the test split is sorted by category; the stratified $N=50$ replication of Appendix C reproduces the headline result.

The scope is open-weight instruction-tuned models of 1.7B to 12B parameters, served through one inference engine. Whether larger or proprietary models share the Llama family’s fragility is untested. Both tasks are English, zero-shot, and objective: a single exactly checkable answer is what lets the parser measure degeneration. The original use case of truncation samplers, open-ended generation with no checkable answer, lies outside this design, so our null result on sampler gains does not extend to it. The recommendation to leave the sampler alone therefore applies to tasks with a checkable answer, served through llama.cpp at the settings above; other engines, other quantization recipes, and open-ended generation are untested. All Llama results come from imatrix GGUFs served by llama.cpp, with no full-precision or second-engine Llama run, so a cause at the level of the conversion recipe or the engine, common to all four quantization levels, is not excluded. Each truncation rule runs at one setting, so the bound applies to those settings rather than to a tuned sampler, and the sampler contrasts were not replicated on the stratified MMLU-Pro subset. Quantization covers llama.cpp k-quants only, one recipe by design so that the quantization axis varies precision and nothing else; other recipes, and other engines’ sampler implementations, may behave differently. Thinking mode is off throughout the main grid; the thinking-on spot-check of Appendix G agrees on GSM8K but shows a 10pp MMLU-Pro drop [2, 20] at $N=25$, so our null does not cover thinking mode.

Every fragile model we observe is a Llama. The lineage run of Section 4.1 shows the fragility spans three generations, which rules out a quirk of one checkpoint. But the three releases share training data and post-training methods, so they are not three independent pieces of evidence. With one fragile family we cannot tell whether the fragility comes from a choice Llama made or from something else these models have in common, and we cannot say how common such families are.

Three analysis caveats. The max-minus-min sampler spread is biased upward, so it overstates the differences between samplers; the bias is conservative for our claim that those differences are small. Accuracy conditional on a strict parse is subject to survivorship bias, which Section 4.3 estimates and corrects. The mechanism probes correlate seven models, and Section 4.6 presents them as consistency checks rather than a causal account. The bound of Section 4.2 is a pointwise 95% upper limit on the sampler effect at $N=50$ (7.8 pp simultaneous), not evidence of no effect, and the grid contains no base-versus-instruct or checkpoint axis, so it can show that instruction tuning does not guarantee robustness but not what moves the cliff. The forced-token experiment uses one configuration, a rank-20 token at position 48 with a 32-position window, and we have not varied position, rank, or number of injections. Mistral’s GSM8K accuracy is low, so its comparison with Llama in Section 4.6 rests on MMLU-Pro.

References

- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. [Training verifiers to solve math word problems](#). *arXiv preprint arXiv:2110.14168*.
- Angela Fan, Mike Lewis, and Yann Dauphin. 2018. [Hierarchical neural story generation](#). In *Proceedings of ACL*.
- Alina Fastowski, Bardh Prenkaj, and Gjergji Kasneci. 2025. [From confidence to collapse in LLM factual robustness](#). *arXiv preprint arXiv:2508.16267*.
- Leo Gao, Jonathan Tow, Baber Abbasi, Stella Biderman, Sid Black, et al. 2023. [A framework for few-shot language model evaluation](#). Zenodo.
- Esteban Garces Arias, Meimingwei Li, Christian Heumann, and Matthias Aßenmacher. 2024. [Decoding decoded: Understanding hyperparameter effects in open-ended text generation](#). *arXiv preprint arXiv:2410.06097*. COLING 2025.
- Gemma Team. 2025. [Gemma 3 technical report](#). *arXiv preprint arXiv:2503.19786*.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, et al. 2024. [The Llama 3 herd of models](#). *arXiv preprint arXiv:2407.21783*.
- Kabir Grover. 2026. [Reliability under randomness: An empirical analysis of sparse and dense language model behavior across decoding temperatures](#). *arXiv preprint arXiv:2601.00942*.

John Hewitt, Christopher D. Manning, and Percy Liang. 2022. Truncation sampling as language model desmoothing . In <i>Findings of EMNLP</i> .	Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-consistency improves chain of thought reasoning in language models . In <i>International Conference on Learning Representations (ICLR)</i> .
Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. 2020. The curious case of neural text degeneration . In <i>International Conference on Learning Representations (ICLR)</i> .	Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, Tianle Li, Max Ku, Kai Wang, Alex Zhuang, Rongqi Fan, Xiang Yue, and Wenhui Chen. 2024. MMLU-Pro: A more robust and challenging multi-task language understanding benchmark . In <i>Advances in Neural Information Processing Systems</i> .
Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. 2023. Mistral 7B . <i>arXiv preprint arXiv:2310.06825</i> .	Sean Welleck, Ilia Kulikov, Jaedeok Kim, Richard Yuanzhe Pang, and Kyunghyun Cho. 2020. Consistency of a recurrent language model with respect to incomplete decoding . In <i>Proceedings of EMNLP</i> .
Clara Meister, Tiago Pimentel, Gian Wiher, and Ryan Cotterell. 2023. Locally typical sampling . <i>Transactions of the Association for Computational Linguistics</i> .	An Yang et al. 2024. Qwen2.5 technical report . <i>arXiv preprint arXiv:2412.15115</i> .
Meta AI. 2024a. Introducing Meta Llama 3: The most capable openly available LLM to date. https://ai.meta.com/blog/meta-llama-3/ .	An Yang et al. 2025. Qwen3 technical report . <i>arXiv preprint arXiv:2505.09388</i> .
Meta AI. 2024b. Llama 3.2: Revolutionizing edge AI and vision with open, customizable models. https://ai.meta.com/blog/llama-3-2-connect-2024-vision-edge-mobile-devices/ .	A Full sampler–temperature grids
Meta AI. 2024c. Meta Llama 3 evaluation details. https://github.com/meta-llama/llama3/blob/main/eval_details.md .	Figure 4 plots the per-model sampler max–min spread of Section 4.2 with its confidence intervals. Tables 4 and 5 give the full main-matrix grids: accuracy for every sampler arm and temperature, per model, pooled over quantization levels.
Minh Nhat Nguyen, Andrew Baker, Clement Neo, Allen Roush, Andreas Kirsch, and Ravid Shwartz-Ziv. 2025. Turning up the heat: Min-p sampling for creative and coherent LLM outputs . In <i>International Conference on Learning Representations (ICLR)</i> .	B Degeneration decomposition
Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. 2019. Language models are unsupervised multitask learners . <i>OpenAI technical report</i> .	Table 7 decomposes accuracy under plain temperature sampling by parse path: the strict final-answer-sentence rate, the parse-failure rate, the fraction of generations that ran to the token cap (including outputs that llama.cpp returned through a parse error, which it raises on degenerate text; the client recovers the text, and the record counts as at cap), and accuracy conditional on a strict parse. Figure 6 shows two representative degenerate outputs. Figure 5 shows one generation per parse path. The strict rule matches the pinned sentence with the answer directly after it, as in the first example. The flexible rule catches two different things: a well-formed answer written outside the pinned sentence, as in the second example, and a derailed generation that still contains a number somewhere in its noise, which GSM8K then grades as a wrong answer. Mistral omits the pinned sentence on 26 to 32% of its GSM8K generations at every temperature, and the Qwen3 models wrap the answer in markdown emphasis, which the strict rule does
Matthew Renze and Erhan Guven. 2024. The effect of sampling temperature on problem solving in large language models . In <i>Findings of EMNLP</i> .	
Rylan Schaeffer, Joshua Kazdan, and Yegor Denisov-Blanch. 2025. Min-p, max exaggeration: A critical analysis of min-p sampling in language models . <i>arXiv preprint arXiv:2506.13681</i> .	
Chufan Shi, Haoran Yang, Deng Cai, Zhisong Zhang, Yifan Wang, Yujiu Yang, and Wai Lam. 2024. A thorough examination of decoding methods in the era of LLMs . In <i>Proceedings of EMNLP</i> .	
Yifan Song, Guoyin Wang, Sujian Li, and Bill Yuchen Lin. 2025. The good, the bad, and the greedy: Evaluation of LLMs should not ignore non-determinism . In <i>Proceedings of NAACL</i> .	
Chenxia Tang, Jianchun Liu, Hongli Xu, and Liusheng Huang. 2024. Top-$n\sigma$: Not all logits are you need . <i>arXiv preprint arXiv:2411.07641</i> .	

Table 4: GSM8K accuracy (%) by sampler and temperature, pooled over quantization.

Model	Sampler	$T=0.7$	$T=1.0$	$T=1.3$
Llama-3.1-8B	greedy ($T=0$)	78.0	–	–
	temperature	78.2	74.8	44.2
	top- k	79.8	75.2	58.7
	top- p	81.5	78.0	63.8
	min- p	81.2	78.0	72.0
	top- $n\sigma$	78.8	80.7	80.8
	top- p (temp-last)	80.8	78.7	73.0
	min- p (temp-last)	78.7	81.3	76.2
Mistral-7B	greedy ($T=0$)	31.0	–	–
	temperature	32.7	30.0	29.2
	top- k	31.8	31.7	29.8
	top- p	28.2	31.0	34.2
	min- p	32.3	28.3	28.8
	top- $n\sigma$	29.5	30.7	30.2
	top- p (temp-last)	29.5	31.2	29.5
	min- p (temp-last)	29.5	32.0	30.5
Qwen2.5-7B	greedy ($T=0$)	93.5	–	–
	temperature	91.5	91.7	87.5
	top- k	90.2	88.5	88.7
	top- p	92.2	91.2	90.3
	min- p	90.8	90.5	91.0
	top- $n\sigma$	92.7	91.2	90.7
	top- p (temp-last)	90.8	91.3	90.3
	min- p (temp-last)	92.0	90.7	91.3
Gemma-3-12B	greedy ($T=0$)	92.0	–	–
	temperature	91.0	90.2	89.0
	top- k	91.8	91.7	90.7
	top- p	91.5	90.3	90.7
	min- p	91.5	91.3	92.3
	top- $n\sigma$	91.3	91.5	91.0
	top- p (temp-last)	90.2	91.8	91.5
	min- p (temp-last)	91.2	90.8	90.3
Qwen3-1.7B	greedy ($T=0$)	75.5	–	–
	temperature	73.3	73.7	73.7
	top- k	72.8	72.7	73.3
	top- p	73.8	75.0	72.7
	min- p	72.2	71.7	74.0
	top- $n\sigma$	74.3	73.5	72.3
	top- p (temp-last)	73.0	72.0	71.3
	min- p (temp-last)	74.7	73.3	74.5
Qwen3-4B	greedy ($T=0$)	95.2	–	–
	temperature	95.2	94.0	93.1
	top- k	94.0	94.5	94.1
	top- p	95.5	95.5	93.5
	min- p	95.1	94.7	92.9
	top- $n\sigma$	94.8	93.9	93.9
	top- p (temp-last)	94.8	95.2	94.3
	min- p (temp-last)	94.5	93.9	93.9
Qwen3-8B	greedy ($T=0$)	94.5	–	–
	temperature	95.0	94.2	94.5
	top- k	94.8	94.8	93.5
	top- p	94.0	94.3	93.7
	min- p	94.8	95.3	94.7
	top- $n\sigma$	96.3	94.2	94.3
	top- p (temp-last)	93.8	94.0	94.0
	min- p (temp-last)	96.2	94.7	95.3

Table 5: MMLU-Pro accuracy (%) by sampler and temperature, pooled over quantization.

Model	Sampler	$T=0.7$	$T=1.0$	$T=1.3$
Llama-3.1-8B	greedy ($T=0$)	54.0	–	–
	temperature	47.8	39.7	7.0
	top- k	51.2	43.3	26.8
	top- p	51.3	46.0	16.0
	min- p	51.2	50.8	39.5
	top- $n\sigma$	51.0	50.5	48.8
	top- p (temp-last)	50.3	46.7	37.5
	min- p (temp-last)	51.2	50.2	43.5
Mistral-7B	greedy ($T=0$)	35.5	–	–
	temperature	30.5	30.2	22.5
	top- k	32.5	32.3	28.3
	top- p	30.2	29.5	25.3
	min- p	33.8	30.7	31.0
	top- $n\sigma$	32.3	31.5	31.5
	top- p (temp-last)	29.2	30.2	32.5
	min- p (temp-last)	33.2	33.3	28.7
Qwen2.5-7B	greedy ($T=0$)	64.0	–	–
	temperature	62.5	61.3	57.7
	top- k	62.5	61.7	57.0
	top- p	65.5	62.8	62.7
	min- p	61.3	63.5	59.7
	top- $n\sigma$	62.0	63.5	61.0
	top- p (temp-last)	63.7	62.7	61.3
	min- p (temp-last)	64.2	63.0	62.3
Gemma-3-12B	greedy ($T=0$)	61.0	–	–
	temperature	63.7	66.7	60.0
	top- k	64.3	62.8	63.7
	top- p	65.7	66.7	63.8
	min- p	65.3	67.0	65.5
	top- $n\sigma$	65.2	65.8	65.8
	top- p (temp-last)	65.8	65.0	65.2
	min- p (temp-last)	65.3	63.7	63.8
Qwen3-1.7B	greedy ($T=0$)	45.5	–	–
	temperature	46.0	45.7	45.3
	top- k	47.5	46.5	45.0
	top- p	47.2	47.0	47.2
	min- p	47.0	47.7	46.2
	top- $n\sigma$	46.8	47.2	46.2
	top- p (temp-last)	47.7	46.8	43.5
	min- p (temp-last)	45.7	46.2	49.0
Qwen3-4B	greedy ($T=0$)	61.6	–	–
	temperature	61.9	62.5	61.2
	top- k	63.5	61.1	61.2
	top- p	62.8	61.3	62.8
	min- p	60.9	62.7	62.9
	top- $n\sigma$	62.4	62.4	61.5
	top- p (temp-last)	62.9	62.7	61.7
	min- p (temp-last)	61.3	63.6	61.5
Qwen3-8B	greedy ($T=0$)	60.5	–	–
	temperature	60.5	59.0	59.5
	top- k	60.2	61.3	57.0
	top- p	60.5	61.3	60.5
	min- p	61.0	59.2	59.7
	top- $n\sigma$	60.0	58.8	58.7
	top- p (temp-last)	60.7	59.5	60.5
	min- p (temp-last)	61.3	60.7	59.3

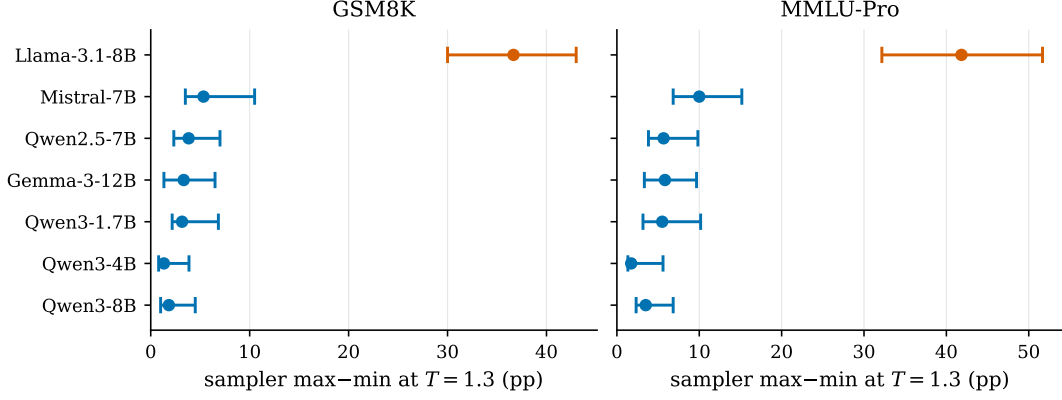


Figure 4: Sampler max-min accuracy spread at $T=1.3$ per model (95% CIs).

[strict] . . . $x = 19.50 / 0.75$ $x = 26$ So the original price of the book was \$26. The final answer is \$26.

[flexible] . . . $80 \text{ \texttt{\text{miles}}} + 150 \text{ \texttt{\text{miles}}} = 230 \text{ \texttt{\text{miles}}}$ ### Final Answer: The final answer is 230.

[failed] . . . $\frac{3,462.20}{35} = 98.92$ ### Final Answer: The answer is J.

Figure 5: One generation per parse path (endings only). Strict: Llama-3.1-8B, GSM8K, $T=1.3$, graded 26, correct. Flexible: Qwen3-8B, GSM8K, $T=0.7$; the emphasis markers defeat the strict rule and the last-number fall-back grades 230, correct. Failed: Qwen3-8B, MMLU-Pro, $T=0.7$; no parenthesized letter anywhere, graded incorrect (the gold answer is E, so the incorrect grade is correct by coincidence).

[opens] To solve this problem, we'll use the gross profit method of inventory evaluation. Step 1: Calculate the gross profit earned during the year. . .

[ends] H sieve Con Palette sediment conduit France called coaching OMAX may PHYS today Enhanced vertices XOR/t and registration ordinances efforts yesterday Michael

[opens] To determine the correct type of research methods, we need to understand the characteristics of each option: A. Non-probability: This method involves selecting. . .

[ends] entageeba lever modeling ign measures inherits cones quantum auditingippines image later campaigns Taylorart parte diesel El N resembled Dental professionals

Figure 6: Two representative degenerate outputs (Llama-3.1-8B, Q8, plain temperature, $T=1.3$, MMLU-Pro; both ran to the 1024-token cap).

not match; both grade correctly through the flexible rule. A failed parse is a generation with no in-range letter or number at all. On MMLU-Pro that is almost always a derailed generation, plus 13 to 16% of Mistral's generations at every temperature that name the option in words rather than by letter, and the markdown case of the third example. Because markdown emphasis defeats the strict rule, we re-graded every record of the main grid with a strict rule that tolerates it: 598 of 185,600 grades change, 556 of them on Qwen3-8B and Qwen3-4B MMLU-Pro; no plain-temperature drop moves by more than 1.0 pp and no deployment-band sampler spread by more than 2.0 pp. The original parser, fixed before any run and mirroring lm-eval-harness, stays primary.

The pattern is consistent: a coherent opening, a mid-generation derailment, then token noise until the cap. The model never reaches an answer at all.

Table 6 gives the length-censoring evidence used

in Section 4.6: run-to-cap rates and the 90th percentile of completion length among *terminated* generations at $T=0.7$ versus $T=1.3$. Robust models are unchanged (ratio 0.85 to 1.19); Llama's surviving generations shorten ($\times 0.73$ GSM8K, $\times 0.55$ MMLU-Pro), so a derailed generation does not recover, and a generation survives only if it finishes before its first bad draw.

C Stratified MMLU-Pro replication

The frozen $N=50$ MMLU-Pro head subset is single-category (business) because the test split is sorted by category. Table 8 repeats the pure-temperature axis (Q8, $K=3$) on a deterministic stratified subset with 3–4 items from each of the 14 categories. The collapse ordering reproduces.

Table 6: Terminated-length censoring under plain temperature at $T=0.7$ versus $T=1.3$ (pooled over quantization). “At cap” includes parse-error-recovered outputs (Appendix B).

Model	Task	at cap (%)		p90 length		ratio
		0.7	1.3	0.7	1.3	
Llama-3.1-8B	GSM8K	5.2	53.8	318	232	$0.73\times$
	MMLU-Pro	11.0	89.2	633	345	$0.55\times$
Mistral-7B	GSM8K	3.5	8.3	367	378	$1.03\times$
	MMLU-Pro	1.0	12.5	628	750	$1.19\times$
Qwen2.5-7B	GSM8K	0.2	1.7	341	361	$1.06\times$
	MMLU-Pro	2.5	10.2	771	732	$0.95\times$
Gemma-3-12B	GSM8K	2.0	5.3	363	344	$0.95\times$
	MMLU-Pro	9.8	17.2	753	763	$1.01\times$
Qwen3-1.7B	GSM8K	2.0	2.2	324	339	$1.05\times$
	MMLU-Pro	15.8	16.3	779	806	$1.03\times$
Qwen3-4B	GSM8K	0.9	0.8	353	361	$1.02\times$
	MMLU-Pro	11.9	12.4	778	731	$0.94\times$
Qwen3-8B	GSM8K	2.3	2.3	379	389	$1.03\times$
	MMLU-Pro	12.3	16.0	850	724	$0.85\times$

D Llama lineage run

The lineage run repeats the pure-temperature axis (greedy anchor plus plain temperature at $T \in \{0.7, 1.0, 1.3\}$) on Llama-3-8B and Llama-3.2-3B: Q8_0, both frozen $N=50$ tasks, $K=3$, 2,000 generations, same pipeline, prompts, and seed derivation as the main matrix. Table 9 reports both models next to the Q8-only Llama-3.1-8B slice of the main matrix, which is why the Llama-3.1 numbers differ slightly from the quantization-pooled Table 2. Greedy anchors are ordinary (Llama-3-8B 76.0% GSM8K / 40.0% MMLU-Pro; Llama-3.2-3B 74.0 / 42.0%). Llama-3.2-3B reproduces the full degeneration signature at $T=1.3$: strict-parse rate $98.7 \rightarrow 41.3\%$ on GSM8K and $82.7 \rightarrow 6.0\%$ on MMLU-Pro, 50 to 51% of generations at the token cap, and accuracy conditional on a strict parse at or above the greedy level (85.5 and 55.6%). Llama-3-8B’s MMLU-Pro drop has a milder surface: strict-parse $97.3 \rightarrow 65.3\%$ with only 4.7% at the cap, so its lost generations mostly terminate as parse failures short of the cap.

E High-temperature regime

Table 10 reports a small probe ($N=15$, $K=3$, Q8) in the regime where min- p and top- $n\sigma$ report their largest gains. The sampler ordering is top- $n\sigma \gg$ min- $p \gg$ top- $p \approx$ temperature; every generation in the zero cells ran to the token cap, and even the most robust model collapses under plain tempera-

ture.

F Temperature-first versus temperature-last

Table 11 gives both chain orders for the two probability-space truncators. The two orders are indistinguishable on the robust models. On Llama at $T=1.3$, truncating before temperature scaling is more protective, worth +9.2 pp on GSM8K and +21.5 pp on MMLU-Pro for top- p (+4.2 and +4.0 for min- p); the results section explains why.

G Thinking-mode spot-check

The matrix runs Qwen3 with thinking off. A confirmatory spot-check (Qwen3-8B, Q8, thinking on, 450 generations, $N=25$, $K=2$) reproduces the GSM8K result (94% at $T=0.7$ versus 96% at $T=1.3$ under plain temperature) but not the MMLU-Pro one: 56% versus 46%, a 10 pp drop with item-clustered CI [2, 20] at $N=25$, with min- p flat on both tasks. With thinking on, the longer traces make Qwen3-8B less robust on the longer task, so the thinking-off null of the main grid should not be read as covering thinking mode.

H Per-quantization-level temperature drops

Figure 7 plots the direct cost of quantization on greedy accuracy. Table 12 recomputes the plain-temperature drop of Table 2 separately at each quantization level. Each cell pools 150 generations ($N=50$, $K=3$), so single-level estimates are noisier than the pooled ones. The BF16 anchor on Qwen3-4B shows the same null drop as its quantized versions (−1.3 pp GSM8K, −2.0 pp MMLU-Pro).

I Answer agreement across repetitions

Mean distinct parsed answers per item across the $K=3$ repetitions under plain temperature (Table 13). Robust models sample diverse token sequences that converge on the same answer; a parse failure counts as one distinct answer, so the statistic is a divergence proxy; it is the per-item agreement that self-consistency decoding (Wang et al., 2023) exploits by majority vote. The parse-failure convention censors Llama-3.1 on MMLU-Pro at $T=1.3$: with 84.5% of generations failing to parse, the repetitions agree on having no answer, and the statistic falls to 1.28. Mistral sits near 2.2 distinct answers at both temperatures, and its scatter has a different

Table 7: Decomposition of the temperature effect (plain temperature, pooled over quantization; %).

Model	Task	T	acc	strict	failed	at cap	acc strict
Llama-3.1-8B	GSM8K	0.7	78.2	93.0	0.0	5.2	83.0
	GSM8K	1.0	74.8	84.7	0.0	7.8	86.4
	GSM8K	1.3	44.2	43.8	0.5	53.8	97.3
	MMLU-Pro	0.7	47.8	82.3	15.3	11.0	58.1
	MMLU-Pro	1.0	39.7	64.8	29.0	23.7	60.2
	MMLU-Pro	1.3	7.0	8.7	84.5	89.2	76.9
Mistral-7B	GSM8K	0.7	32.7	72.2	0.3	3.5	34.2
	GSM8K	1.0	30.0	67.7	0.0	4.3	32.8
	GSM8K	1.3	29.2	62.8	0.0	8.3	29.2
	MMLU-Pro	0.7	30.5	84.0	12.8	1.0	35.7
	MMLU-Pro	1.0	30.2	80.8	17.0	1.0	37.1
	MMLU-Pro	1.3	22.5	73.5	21.2	12.5	29.9
Qwen2.5-7B	GSM8K	0.7	91.5	98.3	0.0	0.2	92.5
	GSM8K	1.0	91.7	95.7	0.0	0.5	92.7
	GSM8K	1.3	87.5	92.5	0.0	1.7	90.1
	MMLU-Pro	0.7	62.5	97.3	2.3	2.5	64.2
	MMLU-Pro	1.0	61.3	96.5	2.5	3.0	63.4
	MMLU-Pro	1.3	57.7	87.3	10.5	10.2	65.6
Gemma-3-12B	GSM8K	0.7	91.0	98.0	0.0	2.0	92.3
	GSM8K	1.0	90.2	96.8	0.0	3.3	93.1
	GSM8K	1.3	89.0	94.5	0.0	5.3	93.8
	MMLU-Pro	0.7	63.7	90.3	9.0	9.8	70.5
	MMLU-Pro	1.0	66.7	89.7	8.2	10.3	74.3
	MMLU-Pro	1.3	60.0	82.8	14.5	17.2	72.2
Qwen3-1.7B	GSM8K	0.7	73.3	20.3	0.0	2.0	73.8
	GSM8K	1.0	73.7	20.0	0.0	0.7	71.7
	GSM8K	1.3	73.7	19.0	0.0	2.2	71.1
	MMLU-Pro	0.7	46.0	83.7	13.3	15.8	54.8
	MMLU-Pro	1.0	45.7	82.2	15.5	16.3	55.4
	MMLU-Pro	1.3	45.3	83.2	13.5	16.3	54.5
Qwen3-4B	GSM8K	0.7	95.2	94.3	0.0	0.9	95.3
	GSM8K	1.0	94.0	92.3	0.0	0.8	94.8
	GSM8K	1.3	93.1	95.6	0.0	0.8	93.7
	MMLU-Pro	0.7	61.9	87.2	12.5	11.9	70.9
	MMLU-Pro	1.0	62.5	87.3	12.0	10.8	71.6
	MMLU-Pro	1.3	61.2	87.5	11.1	12.4	70.0
Qwen3-8B	GSM8K	0.7	95.0	48.2	0.0	2.3	98.3
	GSM8K	1.0	94.2	49.2	0.0	2.3	98.6
	GSM8K	1.3	94.5	52.2	0.0	2.3	98.1
	MMLU-Pro	0.7	60.5	82.5	15.2	12.3	72.3
	MMLU-Pro	1.0	59.0	80.5	15.5	14.8	72.3
	MMLU-Pro	1.3	59.5	79.8	16.3	16.0	73.5

Table 8: Stratified versus head-of-split (business) MMLU-Pro subset; plain temperature accuracy (%) at Q8_0.

Model	Subset	T			
		0.7	1.0	1.3	drop
Llama-3.1-8B	stratified	44.7	38.7	1.3	+43.3 pp
Llama-3.1-8B	head-50	48.0	38.0	10.0	+38.0 pp
Qwen2.5-7B	stratified	54.7	53.3	48.7	+6.0 pp
Qwen2.5-7B	head-50	65.3	66.0	58.0	+7.3 pp
Gemma-3-12B	stratified	64.0	62.7	56.7	+7.3 pp
Gemma-3-12B	head-50	63.3	64.7	60.0	+3.3 pp

source: at a greedy GSM8K accuracy of 31%, most repetitions are wrong, and wrong repetitions rarely agree on the same wrong number at any temperature. The shift attributable to temperature is +0.27.

J Mechanism probes

Figure 8 plots greedy-path entropy against the temperature drop. Table 14 consolidates the per-model mechanism evidence of Section 4.6: greedy-path entropy lower bound and flat-position fraction (top-1 < 0.5) from the summary probe; the temperature-scaled beyond-top-20 escape hazard per position

Table 9: Llama lineage: plain-temperature accuracy and $T_{0.7} \rightarrow 1.3$ drop (pp, 95% item-clustered bootstrap CIs), all at Q8_0. The Llama-3.1-8B rows are the Q8-only slice of the main matrix.

Model	Task	acc (%) at T		drop [95% CI]
		0.7	1.3	
Llama-3-8B	GSM8K	75.3	71.3	+4.0 [−2.7, 11.3]
	MMLU-Pro	44.0	27.3	+16.7 [9.3, 24.0]
Llama-3.1-8B	GSM8K	77.3	47.3	+30.0 [21.3, 38.7]
	MMLU-Pro	48.0	10.0	+38.0 [28.0, 48.7]
Llama-3.2-3B	GSM8K	74.0	36.0	+38.0 [29.3, 47.3]
	MMLU-Pro	40.0	3.3	+36.7 [26.7, 46.0]

Table 10: Accuracy (%) at $T=2$ and $T=3$ (45 observations per cell).

Task	Sampler	Llama-3.1-8B		Gemma-3-12B	
		$T=2$	$T=3$	$T=2$	$T=3$
GSM8K	temperature	0	0	16	0
	top- p	0	0	27	0
	min- p	33	0	78	24
	top- $n\sigma$	71	80	87	89
MMLU-Pro	temperature	0	0	0	0
	top- p	0	0	0	0
	min- p	0	0	58	0
	top- $n\sigma$	47	51	69	62

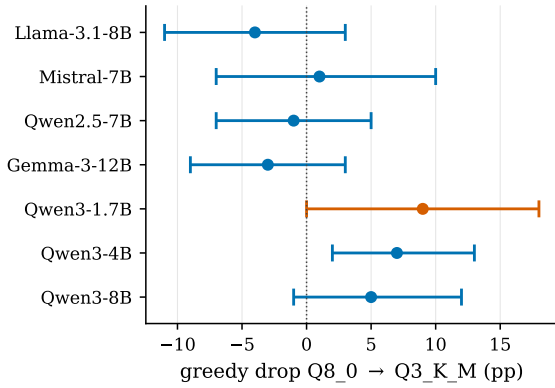


Figure 7: Greedy accuracy drop Q8_0 → Q3_K_M per model (95% CIs), pooled tasks.

at $T=1.3$ from the raw top-20 arrays; and the perturbation deltas (post-injection minus control over 32 positions, escape hazard and entropy) from the forced-token experiment. Escape columns are $\times 10^{-3}$. The escape hazard at temperature T is computed from the raw top-20 probabilities $p_1 \geq \dots \geq p_{20}$ and the residual mass $t = 1 - \sum_i p_i$ as $S_{\text{tail}} / (S_{\text{top}} + S_{\text{tail}})$ with $S_{\text{top}} = \sum_i p_i^{1/T}$; the scaled tail mass is not identifiable from the top 20, so we assume the tail consists of t/p_{20} tokens of probability p_{20} , $S_{\text{tail}} = (t/p_{20}) p_{20}^{1/T}$. The one-token lower

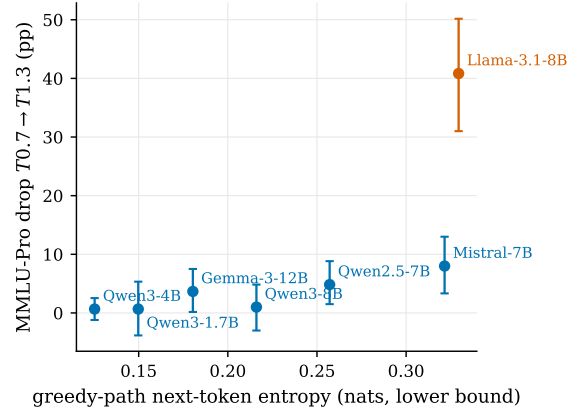


Figure 8: Per-token uncertainty on the model’s own greedy path versus the temperature drop (MMLU-Pro, 95% CIs). The rank correlation is $\rho = +0.93$; the Llama–Mistral gap at equal entropy shows flatness alone does not set the magnitude.

bound $S_{\text{tail}} = t^{1/T}$ ranks the models identically ($\rho = +0.93$). Leaving out any one model, ρ ranges over $[+0.89, +0.94]$ for entropy and for the hazard, and over $[+0.83, +1.00]$ for the perturbation response. Every static column ties Llama-3.1 with Mistral; only the perturbation response separates them. We use the absolute post-injection hazard increase because survival over L tokens scales as $(1-h)^L$ in the absolute hazard h ; it ranks the drops at $\rho = +0.89$, while the ratio of injected to control hazard does not ($\rho = +0.11$), because Gemma-3 and Qwen3-4B triple a hazard that starts far below Llama’s baseline.

K Bound on the sampler effect

Table 15 gives every paired contrast behind Section 4.2: for each model, task, and temperature, the accuracy of each of the six truncation arms minus plain temperature on the same 50 items (each item’s accuracy is its mean over $K=3$ repetitions and the quantization levels, five for Qwen3-4B), with a 95% item-clustered bootstrap CI ($B=5000$). Entries whose interval lies above zero are in bold. A generalized estimating equation over the same records, with an identity link, item clusters, and an exchangeable working correlation, gives the same point estimates and intervals that differ by at most 0.6 pp (median 0.1); scripts `sampler_bound.py` and `sampler_gee.py`.

L Reproducibility

- **Inference stack.** llama.cpp at commit 5aba5364 (server build 9456), native

Table 11: Accuracy (%) with temperature applied before (left) versus after (right) truncation, pooled over quantization.

Model	Method	temperature-first			temperature-last		
		0.7	1.0	1.3	0.7	1.0	1.3
<i>GSM8K</i>							
Llama-3.1-8B	top- <i>p</i>	81.5	78.0	63.8	80.8	78.7	73.0
Llama-3.1-8B	min- <i>p</i>	81.2	78.0	72.0	78.7	81.3	76.2
Mistral-7B	top- <i>p</i>	28.2	31.0	34.2	29.5	31.2	29.5
Mistral-7B	min- <i>p</i>	32.3	28.3	28.8	29.5	32.0	30.5
Qwen2.5-7B	top- <i>p</i>	92.2	91.2	90.3	90.8	91.3	90.3
Qwen2.5-7B	min- <i>p</i>	90.8	90.5	91.0	92.0	90.7	91.3
Gemma-3-12B	top- <i>p</i>	91.5	90.3	90.7	90.2	91.8	91.5
Gemma-3-12B	min- <i>p</i>	91.5	91.3	92.3	91.2	90.8	90.3
Qwen3-1.7B	top- <i>p</i>	73.8	75.0	72.7	73.0	72.0	71.3
Qwen3-1.7B	min- <i>p</i>	72.2	71.7	74.0	74.7	73.3	74.5
Qwen3-4B	top- <i>p</i>	95.5	95.5	93.5	94.8	95.2	94.3
Qwen3-4B	min- <i>p</i>	95.1	94.7	92.9	94.5	93.9	93.9
Qwen3-8B	top- <i>p</i>	94.0	94.3	93.7	93.8	94.0	94.0
Qwen3-8B	min- <i>p</i>	94.8	95.3	94.7	96.2	94.7	95.3
<i>MMLU-Pro</i>							
Llama-3.1-8B	top- <i>p</i>	51.3	46.0	16.0	50.3	46.7	37.5
Llama-3.1-8B	min- <i>p</i>	51.2	50.8	39.5	51.2	50.2	43.5
Mistral-7B	top- <i>p</i>	30.2	29.5	25.3	29.2	30.2	32.5
Mistral-7B	min- <i>p</i>	33.8	30.7	31.0	33.2	33.3	28.7
Qwen2.5-7B	top- <i>p</i>	65.5	62.8	62.7	63.7	62.7	61.3
Qwen2.5-7B	min- <i>p</i>	61.3	63.5	59.7	64.2	63.0	62.3
Gemma-3-12B	top- <i>p</i>	65.7	66.7	63.8	65.8	65.0	65.2
Gemma-3-12B	min- <i>p</i>	65.3	67.0	65.5	65.3	63.7	63.8
Qwen3-1.7B	top- <i>p</i>	47.2	47.0	47.2	47.7	46.8	43.5
Qwen3-1.7B	min- <i>p</i>	47.0	47.7	46.2	45.7	46.2	49.0
Qwen3-4B	top- <i>p</i>	62.8	61.3	62.8	62.9	62.7	61.7
Qwen3-4B	min- <i>p</i>	60.9	62.7	62.9	61.3	63.6	61.5
Qwen3-8B	top- <i>p</i>	60.5	61.3	60.5	60.7	59.5	60.5
Qwen3-8B	min- <i>p</i>	61.0	59.2	59.7	61.3	60.7	59.3

Table 12: Plain-temperature accuracy drop $T_{0.7} \rightarrow 1.3$ (pp) per quantization level.

Model	Q8_0	Q6_K	Q4_K_M	Q3_K_M
<i>GSM8K</i>				
Llama-3.1-8B	+30.0	+32.0	+36.0	+38.0
Mistral-7B	-0.7	-0.7	+3.3	+12.0
Qwen2.5-7B	+4.0	+2.7	+2.7	+6.7
Gemma-3-12B	+3.3	+2.7	+0.0	+2.0
Qwen3-1.7B	+5.3	-6.0	-0.7	+0.0
Qwen3-4B	+2.7	+0.7	+6.0	+2.7
Qwen3-8B	-2.0	+1.3	+2.0	+0.7
<i>MMLU-Pro</i>				
Llama-3.1-8B	+38.0	+46.7	+38.0	+40.7
Mistral-7B	+9.3	+5.3	+10.7	+6.7
Qwen2.5-7B	+7.3	+6.0	+4.0	+2.0
Gemma-3-12B	+3.3	+5.3	+2.7	+3.3
Qwen3-1.7B	+0.7	+6.0	-8.0	+4.0
Qwen3-4B	-0.7	+0.0	+2.0	+4.0
Qwen3-8B	+2.7	-0.7	+0.0	+2.0

Table 13: Mean distinct parsed answers per item ($K=3$, pooled over quantization).

Model	GSM8K		MMLU-Pro	
	$T=0.7$	$T=1.3$	$T=0.7$	$T=1.3$
Llama-3.1-8B	1.51	2.33	1.71	1.28
Mistral-7B	2.21	2.48	1.97	2.21
Qwen2.5-7B	1.10	1.27	1.44	1.57
Gemma-3-12B	1.12	1.16	1.33	1.45
Qwen3-1.7B	1.41	1.43	1.67	1.73
Qwen3-4B	1.05	1.14	1.33	1.38
Qwen3-8B	1.09	1.13	1.41	1.40

tokens, and each record stores the SHA-256 of the exact rendered prompt.

- **Quantizations.** Every quantized checkpoint is a Bartowski imatrix GGUF (one provider, one recipe, so the quantization axis is not confounded by recipe). SHA-256 hashes of all 31 model files are pinned in the repository (DOWNLOADS.md).
- **Decoding.** Exactly one truncation parameter per request; the sampler chain order is pinned

/completion endpoint, CUDA builds of the same commit per GPU architecture (sm_75 for RTX 2080 Ti, sm_120 for RTX 5070 Ti). Server-side BOS insertion is disabled; the client-rendered chat template owns the special

Table 14: Mechanism probes per model (Q8, 100 frozen prompts). Escape hazards are at $T=1.3$; drop is the MMLU-Pro plain-temperature $T_{0.7} \rightarrow 1.3$ accuracy drop (pp), pooled over quantization.

Model	H_{lb}	flat%	esc	Δ_{esc}	ΔH	drop
Llama-3.1-8B	0.329	7.1	3.26	+4.20	+0.110	+40.8
Mistral-7B	0.322	6.8	3.40	+1.19	+0.044	+8.0
Qwen2.5-7B	0.257	4.8	1.97	+1.00	+0.049	+4.8
Gemma-3-12B	0.180	3.0	1.44	+1.42	+0.055	+3.7
Qwen3-1.7B	0.150	2.1	0.36	-0.00	+0.012	+0.7
Qwen3-4B	0.125	1.5	0.22	+0.21	+0.021	+0.7
Qwen3-8B	0.216	3.8	0.71	+0.22	+0.024	+1.0

per arm and recorded in every record. Sampler values: top- p 0.95, top- k 40, min- p 0.05, top- $n\sigma$ 1.0 (each method’s recommended value).

- **Seeding.** Every generation’s seed is derived deterministically from (global seed 0, model, quantization, sampler, temperature, task, item, repetition). Greedy decoding is the deterministic $T=0$ anchor with $K=1$.

- **Tasks.** GSM8K (openai/gsm8k, test split, first 50 items) and MMLU-Pro (TIGER-Lab/MMLU-Pro, test split, first 50 items; stratified replication in Appendix C). Parsers mirror lm-eval-harness; parse failures grade as incorrect and are reported.

- **Records.** Each generation is one JSONL record carrying the full identity tuple, the request parameters, the raw output, the parse path, and throughput covariates; runs are resumable by record id. Batched serving (≤ 8 slots) is seed-logged but not bit-exact.

- **Code and data.** The full pipeline, configs, analysis scripts, and every generation record are in an anonymized repository at <https://anonymous.4open.science/r/decoding-robustness-027C>.

Table 15: Paired accuracy difference of each truncation arm against plain temperature (pp, 95% item-clustered bootstrap CIs; pooled over quantization). Bold: interval above zero.

Model	Task	T	top- p	top- k	min- p	top- $n\sigma$	top- p (T last)	min- p (T last)
Llama-3.1-8B	GSM8K	0.7	+3.3 [+0.2, +6.8]	+1.7 [-1.0, +4.5]	+3.0 [-0.8, +7.0]	+0.7 [-2.7, +4.2]	+2.7 [-0.5, +6.3]	+0.5 [-2.3, +3.2]
		1.0	+3.2 [-0.5, +7.0]	+0.3 [-3.8, +4.3]	+3.2 [-0.7, +7.0]	+5.8 [+2.3, +9.5]	+3.8 [-0.0, +7.8]	+6.5 [+3.0, +10.0]
		1.3	+19.7 [+14.8, +24.5]	+14.5 [+10.0, +19.3]	+27.8 [+22.3, +33.3]	+36.7 [+30.0, +43.3]	+28.8 [+23.3, +34.7]	+32.0 [+25.7, +38.3]
	MMLU-Pro	0.7	+3.5 [+0.3, +6.8]	+3.3 [-0.5, +7.3]	+3.3 [-0.8, +7.7]	+3.2 [-0.2, +6.7]	+2.5 [-1.0, +6.0]	+3.3 [+0.2, +6.7]
		1.0	+6.3 [+2.0, +11.3]	+3.7 [-1.2, +8.5]	+11.2 [+6.7, +16.0]	+10.8 [+6.0, +16.0]	+7.0 [+3.0, +11.2]	+10.5 [+6.2, +14.8]
		1.3	+9.0 [+5.3, +13.3]	+19.8 [+13.5, +26.8]	+32.5 [+24.5, +41.0]	+41.8 [+31.5, +52.3]	+30.5 [+23.0, +38.5]	+36.5 [+27.5, +45.7]
Mistral-7B	GSM8K	0.7	-4.5 [-7.7, -1.3]	-0.8 [-5.7, +3.8]	-0.3 [-3.8, +3.2]	-3.2 [-7.3, +1.2]	-3.2 [-7.5, +1.2]	-3.2 [-6.8, +0.7]
		1.0	+1.0 [-2.7, +4.7]	+1.7 [-1.5, +4.8]	-1.7 [-4.8, +1.7]	+0.7 [-3.0, +4.0]	+1.2 [-3.2, +5.3]	+2.0 [-2.0, +5.8]
		1.3	+5.0 [+0.7, +9.3]	+0.7 [-4.3, +5.5]	-0.3 [-3.8, +3.5]	+1.0 [-2.8, +4.8]	+0.3 [-4.8, +5.0]	+1.3 [-2.3, +5.0]
	MMLU-Pro	0.7	-0.3 [-4.7, +3.8]	+2.0 [-2.3, +6.5]	+3.3 [-0.3, +7.0]	+1.8 [-2.3, +6.2]	-1.3 [-4.7, +2.2]	+2.7 [-1.8, +7.5]
		1.0	-0.7 [-4.7, +3.2]	+2.2 [-1.2, +5.7]	+0.5 [-2.8, +3.8]	+1.3 [-3.2, +5.8]	-0.0 [-4.2, +4.2]	+3.2 [-0.8, +7.3]
		1.3	+2.8 [-1.2, +7.0]	+5.8 [+2.7, +9.2]	+8.5 [+4.5, +12.7]	+9.0 [+4.5, +13.5]	+10.0 [+5.2, +15.2]	+6.2 [+2.2, +10.3]
Qwen2.5-7B	GSM8K	0.7	+0.7 [-1.3, +2.8]	-1.3 [-3.8, +0.8]	-0.7 [-2.0, +0.7]	+1.2 [-0.7, +3.3]	-0.7 [-3.0, +1.5]	+0.5 [-1.2, +2.2]
		1.0	-0.5 [-2.8, +1.7]	-3.2 [-6.0, -0.5]	-1.2 [-3.7, +1.2]	-0.5 [-2.8, +1.7]	-0.3 [-3.3, +2.5]	-1.0 [-3.7, +1.7]
		1.3	+2.8 [+0.7, +5.2]	+1.2 [-0.7, +3.3]	+3.5 [+0.8, +6.3]	+3.2 [+1.2, +5.5]	+2.8 [+0.5, +5.3]	+3.8 [+1.5, +6.5]
	MMLU-Pro	0.7	+3.0 [+0.3, +5.8]	+0.0 [-2.7, +2.7]	-1.2 [-4.5, +2.2]	-0.5 [-3.7, +2.7]	+1.2 [-1.3, +3.5]	+1.7 [-1.2, +4.5]
		1.0	+1.5 [-1.7, +5.0]	+0.3 [-1.8, +2.5]	+2.2 [-1.0, +5.5]	+2.2 [-0.8, +5.3]	+1.3 [-1.0, +3.7]	+1.7 [-1.3, +4.8]
		1.3	+5.0 [+1.7, +8.3]	-0.7 [-3.8, +2.2]	+2.0 [-1.8, +6.2]	+3.3 [-0.3, +6.8]	+3.7 [+0.3, +7.3]	+4.7 [+1.0, +8.7]
Gemma-3-12B	GSM8K	0.7	+0.5 [-1.3, +2.5]	+0.8 [-1.0, +3.2]	+0.5 [-1.3, +2.5]	+0.3 [-1.7, +2.5]	-0.8 [-3.8, +2.0]	+0.2 [-2.2, +2.3]
		1.0	+0.2 [-3.0, +3.2]	+1.5 [-0.5, +3.7]	+1.2 [-1.3, +3.7]	+1.3 [-1.0, +3.7]	+1.7 [-1.0, +4.8]	+0.7 [-1.5, +2.8]
		1.3	+1.7 [-0.2, +3.8]	+1.7 [-0.7, +4.2]	+3.3 [+0.5, +6.7]	+2.0 [-0.0, +4.5]	+2.5 [+0.3, +4.8]	+1.3 [-0.8, +3.5]
	MMLU-Pro	0.7	+2.0 [-0.2, +4.3]	+0.7 [-2.0, +3.3]	+1.7 [-0.8, +4.3]	+1.5 [-1.3, +4.3]	+2.2 [-1.0, +5.5]	+1.7 [-2.2, +5.5]
		1.0	-0.0 [-2.8, +2.8]	-3.8 [-7.2, -0.8]	+0.3 [-2.2, +2.7]	-0.8 [-3.2, +1.3]	-1.7 [-4.8, +1.3]	-3.0 [-6.2, -0.2]
		1.3	+3.8 [+0.3, +8.3]	+3.7 [+0.5, +7.3]	+5.5 [+2.7, +8.7]	+5.8 [+2.2, +10.0]	+5.2 [+2.2, +8.3]	+3.8 [+0.7, +7.3]
Qwen3-8B	GSM8K	0.7	-1.0 [-3.3, +0.8]	-0.2 [-1.5, +1.2]	-0.2 [-1.7, +1.2]	+1.3 [-0.8, +3.5]	-1.2 [-3.5, +0.8]	+1.2 [-0.8, +3.5]
		1.0	+0.2 [-1.7, +2.0]	+0.7 [-1.8, +3.3]	+1.2 [-0.8, +3.3]	-0.0 [-1.8, +1.7]	-0.2 [-2.2, +2.0]	+0.5 [-1.8, +2.8]
		1.3	-0.8 [-2.3, +0.7]	-1.0 [-2.7, +0.5]	+0.2 [-1.8, +2.5]	-0.2 [-1.8, +1.5]	-0.5 [-2.2, +1.0]	+0.8 [-0.3, +2.2]
	MMLU-Pro	0.7	-0.0 [-3.0, +3.2]	-0.3 [-2.7, +2.0]	+0.5 [-2.5, +3.7]	-0.5 [-3.0, +1.8]	+0.2 [-2.7, +2.8]	+0.8 [-2.2, +3.8]
		1.0	+2.3 [-0.7, +5.2]	+2.3 [-0.7, +5.5]	+0.2 [-3.0, +3.5]	-0.2 [-3.0, +2.7]	+0.5 [-2.3, +3.3]	+1.7 [-1.2, +4.7]
		1.3	+1.0 [-1.8, +3.8]	-2.5 [-5.8, +0.7]	+0.2 [-3.2, +3.3]	-0.8 [-4.2, +2.3]	+1.0 [-1.8, +3.8]	-0.2 [-3.3, +3.2]
Qwen3-4B	GSM8K	0.7	+0.3 [-0.9, +1.6]	-1.2 [-2.9, +0.7]	-0.1 [-1.3, +1.3]	-0.4 [-1.6, +0.9]	-0.4 [-2.0, +1.6]	-0.7 [-2.5, +1.6]
		1.0	+1.5 [+0.0, +3.1]	+0.5 [-0.7, +1.7]	+0.7 [-1.1, +2.7]	-0.1 [-1.5, +1.1]	+1.2 [+0.0, +2.4]	-0.1 [-1.5, +1.3]
		1.3	+0.4 [-1.2, +2.0]	+1.1 [-0.7, +2.9]	-0.1 [-1.9, +1.6]	+0.8 [-1.1, +2.7]	+1.2 [-0.5, +3.1]	+0.8 [-0.7, +2.5]
	MMLU-Pro	0.7	+0.9 [-1.6, +3.2]	+1.6 [-0.5, +3.9]	-0.9 [-3.6, +1.5]	+0.5 [-1.6, +2.5]	+1.1 [-0.9, +3.2]	-0.5 [-3.1, +1.9]
		1.0	-1.2 [-3.2, +0.8]	-1.5 [-3.5, +0.4]	+0.1 [-2.7, +3.2]	-0.1 [-2.9, +2.4]	+0.1 [-2.8, +3.2]	+1.1 [-1.5, +3.6]
		1.3	+1.6 [-1.1, +4.5]	+0.0 [-2.8, +2.5]	+1.7 [-0.7, +4.1]	+0.3 [-2.3, +2.7]	+0.5 [-2.1, +3.2]	+0.3 [-2.3, +2.8]
Qwen3-1.7B	GSM8K	0.7	+0.5 [-2.8, +3.8]	-0.5 [-3.8, +2.8]	-1.2 [-4.7, +2.3]	+1.0 [-2.3, +4.3]	-0.3 [-3.0, +2.3]	+1.3 [-1.5, +4.3]
		1.0	+1.3 [-2.0, +4.5]	-1.0 [-4.2, +2.0]	-2.0 [-4.3, +0.2]	-0.2 [-2.7, +2.3]	-1.7 [-4.2, +0.7]	-0.3 [-3.2, +2.5]
		1.3	-1.0 [-4.5, +2.2]	-0.3 [-3.3, +2.8]	+0.3 [-3.0, +3.7]	-1.3 [-4.2, +1.2]	-2.3 [-6.0, +1.2]	+0.8 [-2.2, +3.7]
	MMLU-Pro	0.7	+1.2 [-1.8, +4.2]	+1.5 [-1.7, +4.7]	+1.0 [-2.5, +4.7]	+0.8 [-2.7, +4.2]	+1.7 [-1.3, +4.8]	-0.3 [-3.5, +2.8]
		1.0	+1.3 [-2.5, +5.0]	+0.8 [-2.3, +4.2]	+2.0 [-1.5, +5.3]	+1.5 [-2.0, +4.7]	+1.2 [-2.3, +4.8]	+0.5 [-3.3, +4.2]
		1.3	+1.8 [-2.8, +6.7]	-0.3 [-4.2, +3.5]	+0.8 [-3.8, +5.5]	+0.8 [-3.5, +5.2]	-1.8 [-6.3, +2.7]	+3.7 [-0.8, +8.3]